ISPDiff: Interpretable Scale-Propelled Diffusion Model for Hyperspectral Image Super-Resolution

Wenqian Dong, *Member, IEEE*, Sen Liu, Song Xiao, Jiahui Qu, *Member, IEEE*, and Yunsong Li, *Member, IEEE*

***Abstract*— Hyperspectral image (HSI) super-resolution (SR) employing the denoising diffusion probabilistic model (DDPM) holds significant promise with its remarkable performance. However, existing relevant works exhibit two limitations: 1) directly applying DDPM to fusion-based HSI-SR ignores the physical mechanism of HSI-SR and unique characteristics of HSI, resulting in less interpretability and 2) scale-invariant DDPM suffers from a time-consuming inference. To tackle these issues, we propose an interpretable scale-propelled diffusion (ISPDiff) model for HSI-SR, which combines the underlying principles of HSI-SR with DDPM for progressively unrolling reconstruction by learning its distribution at various scales, enhancing the transparency significantly and reducing the inference time prominently. Concretely, we destroy and downsample HSI into Gaussian noise in the forward process of ISPDiff. Then we design a unified scale-flexible model in the backward process to iteratively refine HSI in a coarse-to-fine manner through scale-matched reconstruction and cross-scale upsampling, which can be unfolded with optimization algorithms. These solved equations are one-to-one corresponding unrolled into two deep neural networks, called progressive perceptual model-driven scale-matched restoration network (P^2MSRN) and cross-scale model-driven upsampling network (CMUN). Through end-to-end training, the proposed ISPDiff implements HSI-SR with a scale-propelled unrolling diffusion characterized by enhanced interpretability, stronger task orientation, and reduced time consumption. Systematic experiments have been conducted on three public datasets, demonstrating that ISPDiff outperforms state-of-the-art methods. The code is available at https://github.com/Jiahuiqu/ISPDiff.**

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***Index Terms*— Deep unrolling network, diffusion model, hyperspectral image (HSI), super-resolution (SR).**

I. Introduction

HYPERSPECTRAL image (HSI) [1] with rich spectral information obtained by optical remote sensing satellites plays a crucial role in many tasks, including change detection [2], [3], classification [4], [5], and anomaly detection [6]. However, owing to the inherent constraints of imaging hardware, it is challenging to acquire data that possesses both high spectral and spatial resolutions simultaneously. When it comes to practical applications, solely relying on HSI falls short of meeting real-world requirements. As a result, many HSI super-resolution (SR) algorithms have been proposed to tackle this problem, which can be categorized into single-image SR and fusion-based SR.

Single-image SR refers to the process of reconstructing high-resolution images from one low-resolution input, which mainly focuses on learning the mapping between the low-resolution and high-resolution images to predict the missing high-frequency details. For example, Hu et al. [7] propose a residual structure-textual dense network to exploit the important observation that the structure and the texture exhibit different sensitivities to the spatial resolution, opening a new way for superresolving the HSI from a structure and texture recovery aspect. However, it is important to note that this approach usually faces limitations in capturing the authentic details that correspond to the high-resolution image. Fusion-based HSI-SR aims to fuse multiple registered low-resolution images to generate high-resolution ones, as shown in Fig. 1. Auxiliary images involved in the HSI-SR, such as RGB image and multispectral image (MSI) [8] which contain less spectral band but more spatial details, could help to enhance the quality and fidelity of the HSI. Therefore, HSI-SR algorithms use the information from multiple images and generally yield reconstructed high-resolution HSIs (HrHSIs) that are more accurate and of higher quality compared to single HSI SR methods. Li et al. [9] introduce CasFormer, a novel cascaded transformer architecture tailored for fusion-aware computational hyperspectral imaging, which enables effective enhancement of hyperspectral imaging quality by merging RGB images, with particular emphasis on spatial and spectral domains. Li et al. [10] propose a deep architec-

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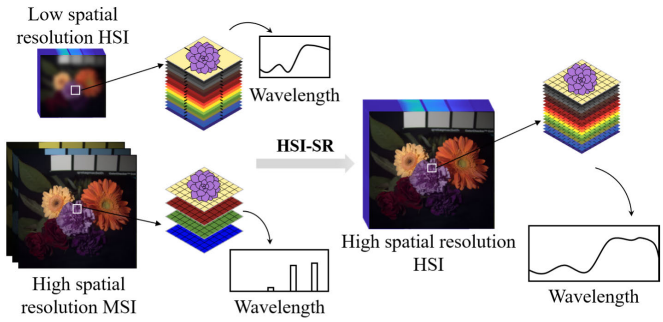


Fig. 1. Schematic of fusion-based HSI-SR.

ture using a single scale factor in terms of the symmetrical guided encoder to explore the HSI and RGB image, taking advantage of the inherent priors of spatial and spectral information.

The exponential development of deep learning (DL) has propelled advancements across various fields, and the one of remote sensing [11], [12] has also benefited greatly. The application of neural networks, notably CNNs, Transformers, variational autoencoders (VAEs), and generative adversarial networks (GANs), in SR, has been relatively mature, while typically these models still encounter common and formidable challenges. CNNs [13], [14] exhibit excellent reconstruction performance, however, its internal structure often lacks transparency and theoretical support. As for Transformers, though they are computationally efficient and adept at handling longer sequence data, they entail high model complexity and necessitate substantial training data. Li et al. [15] propose a dual-stage learning approach for HSI SR to learn general spatial–spectral prior and image-specific details to rich spectra of HSI and HR content of RGB image. The VAE-based models [16] often yield suboptimal quality, especially when confronted with intricate images, which need an elaborate endeavor to train. The GAN-based [17] models have demonstrated their ability to generate results with high quality, yet they are plagued by issues of instability to train, lack of interpretability, and susceptibility to pattern collapse, which significantly impact the overall quality of the super-resolved HSI. Liu et al. [18] design a multiscale unsupervised architecture based on GANs to produce high-resolution MSI, which consists of a generator and spectra/spatial discriminators used against the generator.

As a new powerful generative model, the denoising diffusion probabilistic model (DDPM) is well-known for its high-quality generation, strong interpretability, and high flexibility. Therefore, several published works have exploited the effectiveness of DDPM in HSI-SR [19], which can be a solution to alleviate the dilemmas of the aforementioned models. For example, Li et al. [20] introduce DCDM comprising two noise prediction networks with distinct conditional inputs, tailored to extract image features from the low-resolution HSI (LrHSI) and panchromatic images, accommodating the varied characteristics of the input images and reconstructing the HRHSI features for fusion. However, these methods directly combine DDPM with HSI-SR at one singular scale without taking into account the practical characteristics of HSIs. In addition, these networks are usually designed as black boxes, which might not be inherently explainable.

In this article, a novel interpretable scale-propelled diffusion (ISPDiff) model for HSI-SR named ISPDiff is proposed. ISPDiff focuses on reconstructing HrHSI by capturing hierarchical details and structures across various scales in an understandable way. ISPDiff degrades and downsamples HrHSI to Gaussian noise of the same size as the LrHSI in the forward process. In the reverse process, to prevent the loss of information and the introduction of artifacts and to enhance the interpretability of the model, we reconstruct LrHSI based on the physical principles of HSI-SR at multiple scales. Precisely, we iteratively refine the corrupted HSI from coarse to fine through scale-matched reconstruction and cross-scale upsampling. These two operations can be uniformly formulated into a transparent unified model, which can be optimized by iterative algorithms and one-to-one corresponding unfolded into two homologous networks: progressive perceptual model-driven scale-matched restoration network (P^2MSRN) and cross-scale model-driven upsampling network (CMUN). As we perform denoising across various scales in the backward, P^2MSRN undertakes the restoration of HSI for a specific amount of time steps at a low level, subsequently transfers the intermediate result to CMUN for upsampling, which afterward triggers P^2MSRN to reconstruct HSI at a higher level. To summarize, the contributions of this work are as follows.

1) In this article, we present ISPDiff, a novel interpretable scale-driven model for HSI SR. ISPDiff stands out by seamlessly integrating features from diverse scales in a transparent and understandable manner, constructing a coarse-to-fine reconstruction process, which not only emphasizes interpretability but also delivers high-quality reconstructions.
2) We design a unified scale-flexible model in the backward process across various scales based on the observation model to reconstruct the latent HSI, which can be one-to-one corresponding solved and unfolded into networks: P^2MSRN for reconstruction and CMUN for upsampling. This design can make the reconstruction and upsampling operation both have specific physical meanings.
3) We perform interpretable diffusion on multiple scales based on mathematical models, which makes full use of information from two sources at different scales in an explainable way, and this strategy largely contributes to reducing the time required for the inference.

The remainder of the article is structured as follows. Section II provides a review of related works. In Section III, we introduce the proposed ISPDiff for HSI SR. Section IV presents qualitative and quantitative results on three public datasets and one real dataset, along with ablation studies to validate specific design choices. Finally, Section V concludes this article.

II. Related Works

In this section, we review two respective lines of research related to this article, encompassing both existing HSI-SR methodologies, advancements of DDPM, and the development of the unrolling technique in DL.

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A. HSI-SR Methods

Presently, HSI-SR methods can be mainly divided into four categories: 1) component substitution (CS); 2) multiresolution analysis (MRA); 3) model-based methods; and 4) DL-based. Yokoya et al. [21] propose that hyperspectral and multispectral data are alternately unmixed into end member and abundance matrices by the CNMF algorithm based on a linear spectral mixture model. Selva et al. [22] achieve the generation of a fused HSI by directly integrating the spatial details of an MSI into the HSI counterpart. Lin et al. [23] propose coupled nonnegative matrix factorization along with a sum-of-squared-distances regularizer. This approach alternately updates the spectral basis and coefficients, resulting in a robust fusion performance. The first two types of traditional approaches encounter the problems of significant spectral distortion and model-based methods [24] heavily rely on manually designed prior assumptions. To overcome the above difficulties, DL-based methods [10] are developed. Zhang et al. [25] propose to learn a deep image-specific prior for each HSI. Instead of directly modeling a complicated prior, they utilize deep networks to implicitly learn a general image prior that applies to most HSIs. FusionNet [26], a novel VAE framework, integrates spatial and spectral information from LrHSI and high-resolution MSI (HrMSI). Enhanced by meta-learning, it enables rapid adaptation to diverse scenes. Li et al. [27] explore the discovery that HSI exhibits high similarity between adjacent bands in a large spectral range, therefore they propose a novel dual-stage structure to recover the HSI from coarse to fine. Interactformer [28] employs 3-D CNN and Transformer for extracting local and global features respectively and incorporates an interactive attention mechanism to establish connections between these two features, resulting in an enhanced spatial resolution HSI. Li et al. [29] propose an RGB-induced feature modulation network for HSI SR by utilizing the abundant high-resolution RGB details as a priori information to enhance the HSI feature representation, improving the overall quality of hyperspectral data. Dong et al. [30] propose a generative model for HSI-SR with one generator and two discriminators to maintain both the spectral and spatial information simultaneously, but it inherits the challenge of unstable GAN training. Although these models deliver satisfactory results, they are based on a "black-box" design.

B. Denoising Diffusion Probabilistic Model

DDPM [31] has emerged as the new state-of-the-art family of deep generative models using a Markov chain to transform latent variables into simple distributions, which has been introduced to multiple tasks. SR3 [32] proposed by Google Research to adopt DDPM to conditional image generation by adding extra images to the inference process. Due to the existing computational burden of DDPM, Rombach et al. [33] propose LDM to free the DDPM from the enormous need for computational resources by introducing DDPM into the latent space. One advantage of DDPM is its ability to capture the complex statistics of the visual world. Thus, we have incorporated the diffusion model into the HSI-SR task, namely ISPDiff. Existing approaches have roughly extended the DDPM to HSI-SR without considering the unique characteristics of HSI, and they face the difficulty of time-consuming reference. To address these limitations, we design an ISPDiff model specifically for HSI-SR.

C. Unrolling Technique

The advancements in DL have promoted the swift evolution of HSI fusion, with researchers continually introducing novel network architectures to improve the quality of fused HSIs. However, most of these methodologies are characterized by "black box" models, devoid of interpretability, thereby challenging their applicability in real-world scenarios. To enhance the transparency in these deep networks, some researchers have introduced the unfolding techniques into HSI fusion [34], anomaly detection [35], and classification [36]. The essence of unfolding networks lies in deriving their fundamental structure from the iterative optimization process of model-based algorithms, with trainable parameters automatically acquired through the training process. Essentially, these techniques involve mapping the target task onto iterative optimization solvers. Subsequently, each iteration can be translated into a network operation with practically physical meaning, thereby constructing the network by concatenating multiple iterations or layers. Then, this operation would provide the network with significant interpretability.

III. Methods

In this section, we will introduce the proposed ISPDiff in detail, which models upsampling and reconstruction processes uniformly. The model can be solved through an optimization algorithm and one-to-one corresponding unfolded into two respective networks, namely CMUN and P^{2}MSRN, each of which consists of the same network structures, denoted as TRNet and InDNet. Following multiple time steps of reconstruction and upsampling in the diffusion model, we ultimately obtain the desired HrHSI from coarse to fine.

A. Model Formulation

The existing approaches for HSI-SR based on DDPM take HrMSI as an auxiliary condition and directly upsample LrHSI to the same size as the HrMSI for reconstruction. However, this single-scale and black-box approach does not consider the characteristics of HSI and lacks task dependency, resulting in reconstruction artifacts and loss of detail. Therefore, ISPDiff reconstructs HrHSI by injecting HrMSI with the corresponding resolution into LrHSI level by level in an interpretable way. Specifically, while adding noise to HrHSI in the forward process, we also downsample it with a ratio of factor f_t at the tth step. In the reverse process, we spend a certain amount of time steps to reconstruct the corrupted HSI on a constant scale guided by HrMSI. Then, ISPDiff performs upsampling with a ratio of $1/f_t$ on this partially reconstructed HSI to facilitate a subsequent stage of restoration at a larger scale. This sequential reconstruction on multiple scales can develop a continuous memory mechanism on a consistent scale by iteratively refining the result of the former step.

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In detail, ISPDiff consists of two stages: scale-diminished diffusion (forward) and scale-propelled denoising (backward) processes. As for the forward process, we add Gaussian noise into HrHSI $\mathbf{X_0}$ while downsampling $\mathbf{X_0}$ with factor f_t. It can be formulated as follows:

$$q(\mathbf{X}_t|\mathbf{X}_{t-1}) = N(\mathbf{X}_t; \sqrt{\alpha_t}(\mathbf{X}_t \downarrow_{f_t}), (1-\alpha_t)\mathbf{I}) \tag{1}$$

where α_t is the noise schedule, f_t is the downsampling factor, and $\mathbf{I}$ is an identity matrix. Accordingly, ISPDiff proceeds with the scale-propelled denoising as follows:

$$p_\theta(\mathbf{X}_{t-1}|\mathbf{X}_t) = N\left(\mu_\theta\left(\mathbf{X}_t \uparrow_{1/f_t}, t\right), \sigma_t^2 \mathbf{I}\right) \tag{2}$$

where θ is the learnable parameters that can be acquired through network design in the reverse process, μ_θ and σ_t^2 denote the mean and variance estimated by the network at tth step.

Considering that conducting successive HSI-SR at various scales facilitates the acquisition of spatial relationships and structures at each level, we devise a scale-propelled diffusion. Meanwhile, to promote a better comprehension of HSI-SR and bolster the reliability of the model, we focus on ensuring that the reverse process is deeply interpretable for HSI-SR, instead of indiscriminately constructing a complicated deep network. Concretely, we formulate a transparent unified scale-flexible model for two distinct physical models: scale-matched reconstruction and cross-scale upsampling, which can be represented as follows:

$$\mathbf{Z}_i = \begin{cases} \mathbf{X}_i(\uparrow_{1/f_i})\mathbf{B}_i + \mathbf{N}_H, & f_i = f_{i-1} \quad (3a) \\ \mathbf{X}_i(\uparrow_{1/f_i})\mathbf{S}_i + \mathbf{N}_H, & f_i \neq f_{i-1} \quad (3b) \end{cases}$$

$$\mathbf{Y}_i = \mathbf{R}_i\mathbf{X}_i(\uparrow_{1/f_i}) + \mathbf{N}_M \tag{4}$$

where i denotes the ith time step in the backward process, $\mathbf{X}_i \in \mathbb{R}^{L\times M}$ is the desired HrHSI with L bands and M pixels, $\mathbf{Z}_i \in \mathbb{R}^{L\times m}$ and $\mathbf{Y}_i \in \mathbb{R}^{l\times M}(l \ll L, m \ll M)$ represent the observed LrHSI and HrMSI, respectively, $\mathbf{B}_i$ is the blurring matrix, $\mathbf{S}_i$ is the blurring and downsampling matrix, $\mathbf{R}_i$ is the spectral response, and $\mathbf{N}_H$ and $\mathbf{N}_M$ are the noise contained in LrHSI and HrMSI.

At scale k, all time steps aim to be scale-matched reconstruction, as modeled by (3a) and (4), furthering iterative refinement at the identical scale (i.e., $f_i = f_{i-1}$). Since this model describes the process of acquiring HrMSI and only-blurred LrHSI, the refinement is conditioned on both LrHSI and HrMSI with the same resolution. Upon the need for restoration at a finer scale $k+1$, we adopt the model defined by (3b) and (4) to perform cross-scale upsampling when $f_i \neq f_{i-1}$, ensuring an understandable transition to a higher scale without incorporating erroneous details. This model illustrates the procedure for obtaining HrMSI and downsampled LrHSI, utilizing LrHSI and higher-scale HrMSI as conditional inputs to yield LrHSI with higher spatial resolution. The overall process can be briefly illustrated in Fig. 2. This interleaved reconstruction and upsampling operation not only captures the structural and detailed information but also shortens the inference. These two models can be unfolded in an analogical way as shown below.

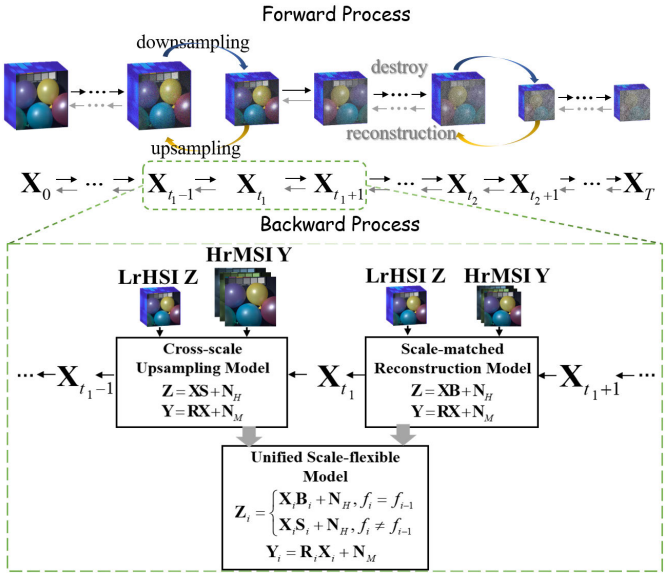


Fig. 2. Overall diagram of the proposed ISPDiff.

B. Optimization Algorithm

1) Scale-Matched Reconstruction Model Optimization: From (3a) to (4), it is necessary to incorporate prior knowledge of the desired HrHSI through the utilization of regularization techniques. The objective function is formulated as follows:

$$\mathbf{X}_i^* = \arg\min_{\mathbf{X}} \frac{1}{2}||\mathbf{Z}_i - \mathbf{X}_i\mathbf{B}_i||_F^2 + \frac{1}{2}||\mathbf{Y}_i - \mathbf{R}_i\mathbf{X}_i||_F^2 + \lambda_1\phi(\mathbf{X}_i) \tag{5}$$

where the first and second terms in the equation correspond to fidelity terms, the last one represents a regularization term, λ denotes the Lagrangian multiplier, the Frobenius norm is denoted by $||\cdot||_F$, and $\phi(\,)$ is a prior term.

We have opted for the half-quadratic splitting method as the optimization algorithm to solve (5), considering that it can handle high-dimensional data while maintaining high accuracy and computational efficiency compared to other algorithms. By introducing an auxiliary splitting variable G, we can reformulate (5) as an optimization problem

$$(\mathbf{X}_i^*, \mathbf{G}_i^*) = \arg\min_{\mathbf{X},\mathbf{G}} \frac{1}{2}||\mathbf{Z}_i - \mathbf{X}_i\mathbf{B}_i||_F^2 + \frac{1}{2}||\mathbf{Y}_i - \mathbf{R}_i\mathbf{X}_i||_F^2 + \lambda_1\phi(\mathbf{X}_i), \quad \text{s.t. } \mathbf{X}_i = \mathbf{G}_i \tag{6}$$

and the equally constrained optimization problem can be converted into an unconstrained optimization problem

$$\mathbf{X}_{i_{k+1}}^* = \arg\min_{\mathbf{X}} \frac{1}{2}||\mathbf{Z}_i - \mathbf{X}_i\mathbf{B}_i||_F^2 + \frac{1}{2}||\mathbf{Y}_i - \mathbf{R}_i\mathbf{X}_i||_F^2 + \beta_1||\mathbf{G}_{i_k} - \mathbf{X}_i||_F^2 \tag{7}$$

$$\mathbf{G}_{i_{k+1}}^* = \arg\min_{\mathbf{G}} \beta_1||\mathbf{G}_i - \mathbf{X}_{i_{k+1}}||_F^2 + \lambda_1\phi(\mathbf{G}_i). \tag{8}$$

By applying the Gradient Descent Algorithm and the properties of Frobenius norm, we can turn (7) into the following

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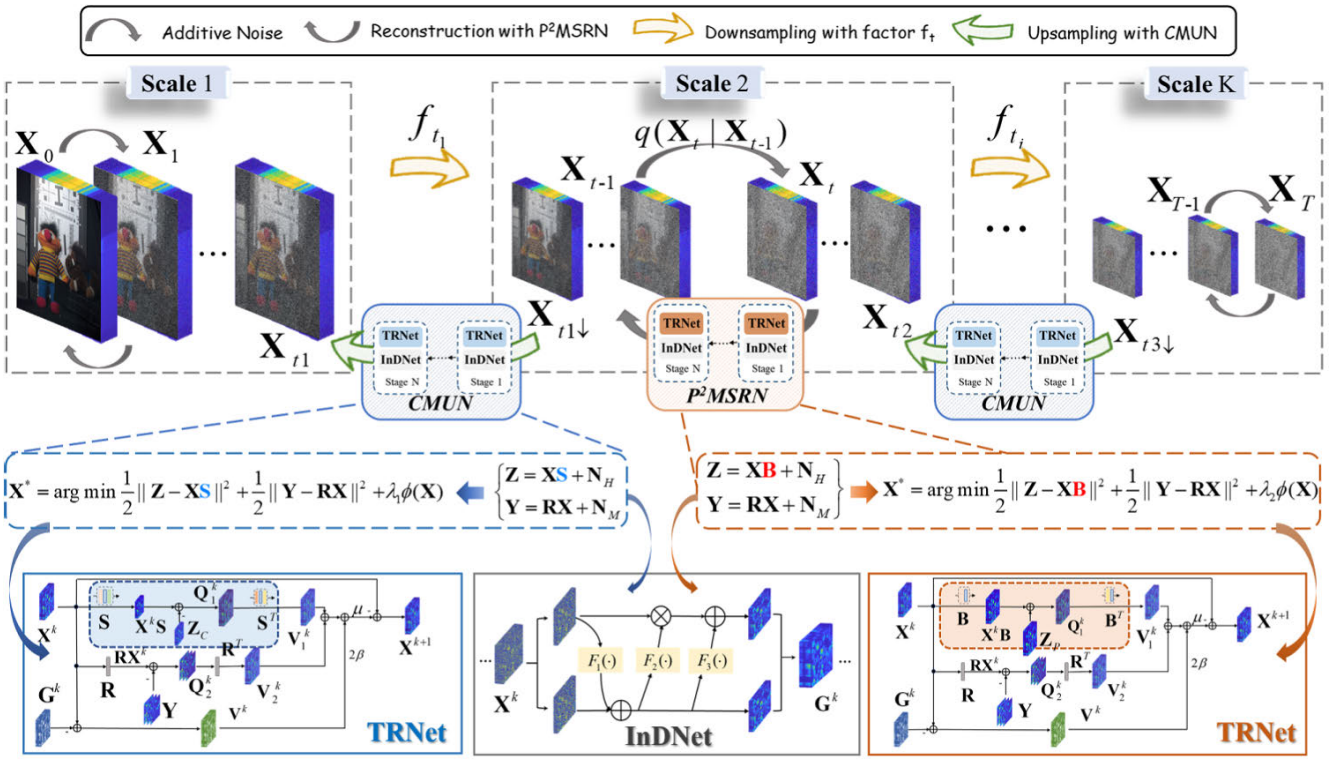


Fig. 3. Overall structure of ISPDiff. In the forward process, we not only introduce Gaussian noise into HrHSI $\mathbf{X}_0$ but implement downsampling with a ratio f_t after certain steps, achieving diffusion on K scales. During the restoration across multiple scales in the reverse process, P^{2}MSRN and CMUN for scale-matched reconstruction and cross-scale upsampling are founded on similar mathematical models and can be unfolded into the same structure (i.e., TRNet and InDNet) with distinct inputs. The desired HrHSI can be delivered in such a way that the reconstruction and upsampling are performed alternately.

formula:

$$\nabla_{\mathbf{X}} f(\mathbf{X}_i) = \mathbf{X}_i^{(k)}\mathbf{B}_i\mathbf{B}_i^T + \mathbf{R}_i^T\mathbf{R}_i\mathbf{X}_i^{(k)} - \mathbf{Z}_i\mathbf{B}_i^T - \mathbf{R}_i^T\mathbf{Y}_i + 2\beta_1\mathbf{X}_i^{(k)} - 2\beta_1\mathbf{G}_i^{(k)}. \tag{9}$$

Therefore, the solution to the $(k+1)$th stage of $\mathbf{X}_i$ in the scale-matched reconstruction model can be signified as follows:

$$\mathbf{X}_{i_{k+1}} = \mathbf{X}_{i_k} - \mu_1\left[(\mathbf{X}_{i_k}\mathbf{B}_i - \mathbf{Z}_i)\mathbf{B}_i^T + \mathbf{R}_i^T(\mathbf{R}_i\mathbf{X}_{i_k} - \mathbf{Y}_i) + 2\beta_1(\mathbf{X}_{i_k} - \mathbf{G}_{i_k})\right] \tag{10}$$

where μ_1 is a learnable parameter.

2) Cross-Scale Upsampling Model Optimization: Similarly, the half quadratic splitting method is utilized to optimize the cross-scale upsampling model [i.e., (3b) and (4)], which can be transformed into the following constrained optimization problem:

$$(\mathbf{X}_i^*, \mathbf{G}_i^*) = \arg\min_{\mathbf{X},\mathbf{G}} \frac{1}{2}||\mathbf{Z}_i - \mathbf{X}_i\mathbf{S}_i||_F^2 + \frac{1}{2}||\mathbf{Y}_i - \mathbf{R}_i\mathbf{X}_i||_F^2 + \lambda_2\phi(\mathbf{X}_i), \quad \text{s.t. } \mathbf{X}_i = \mathbf{G}_i \tag{11}$$

where λ_2 is a learnable parameter. By decoupling the fidelity and regularization terms, we also obtain the following iterative equations:

$$\mathbf{X}_{i_{k+1}}^* = \mathbf{X}_{i_k} - \mu_2\left[(\mathbf{X}_{i_k}\mathbf{S}_i - \mathbf{Z}_i)\mathbf{S}_i^T + \mathbf{R}_i^T(\mathbf{R}_i\mathbf{X}_{i_k} - \mathbf{Y}_i) + 2\beta_2(\mathbf{X}_{i_k} - \mathbf{G}_{i_k})\right] \tag{12}$$

$$\mathbf{G}_{i_{k+1}}^* = \arg\min_{\mathbf{G}} \beta_2||\mathbf{G}_i - \mathbf{X}_{i_{k+1}}||_F^2 + \lambda_2\phi(\mathbf{G}_i). \tag{13}$$

C. Progressive Perceptual Model-Driven Scale-Matched Restoration Network

With the solution of (8) and (10), an unfolded network named P^{2}MSRN, which is employed when ISPDiff performs reconstruction on the consistent scale, can be designed. P^{2}MSRN comprises a composition of the following two networks.

1) Transparent Reconstruction Network: Given (10), it can be unrolled into a transparent reconstruction network (TRNet) to compute $\mathbf{X}_i^{k+1}$ iteratively. We can decompose this updating rule into the following equations:

$$\begin{cases} \mathbf{Q}_{i_1}^k = \mathbf{X}_i^k\mathbf{B}_i - \mathbf{Z}_i \\ \mathbf{V}_{i_1}^k = \mathbf{Q}_{i_1}^k\mathbf{B}_i^T \end{cases} \tag{14}$$

$$\begin{cases} \mathbf{Q}_{i_2}^k = \mathbf{R}_i\mathbf{X}_i^k - \mathbf{Y}_i \\ \mathbf{V}_{i_2}^k = \mathbf{R}_i^T\mathbf{Q}_{i_2}^k \end{cases} \tag{15}$$

$$\mathbf{V}_i{}^k = \mathbf{X}_i{}^k - \mathbf{G}_i{}^k \tag{16}$$

$$\mathbf{X}_i{}^{k+1} = \mathbf{X}_i{}^k - \mu_1\left(\mathbf{V}_{i_1}^k + \mathbf{V}_{i_2}^k + 2\beta_1\mathbf{V}_i{}^k\right) \tag{17}$$

where (14) updates the spectral information in each stage, $\mathbf{X}_i^k\mathbf{B}_i - \mathbf{Z}_i$ represents the difference of spectrum between the available LrHSI and spatially blurred version of the output by the kth iteration, and (15) denotes the spatial information update, where $\mathbf{R}_i\mathbf{X}_i^k - \mathbf{Y}_i$ shows the spatial difference.

As shown in Fig. 3, each operator of TRNet in P^{2}MSRN has a one-to-one correspondence with the equations, preserving the original physical meaning. To TRNet in P^{2}MSRN, LrHSI and

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HrMSI with the same size serve as the inputs, along with the output $\mathbf{X}_i^k$ and $\mathbf{G}_i^k$ from the previous stage. Besides, following the scale-matched reconstruction model, $\mathbf{X}_i^k$ will pass through P^{2}MSRN branch in TRNet, where $\mathbf{B}_i$ is equivalent to the blurring matrix. All the variables in (14) and (15) can be translated into operations in CNN, which enables each module to maintain its inherent physical meaning. $\mathbf{B}_i$ can be converted into a sequential which consists of a 3×3 Conv and a PReLU for blurring and activation, respectively. In detail, $\mathbf{B}_i^T$ is comprised of 3×3 Deconv and PReLU. In addition, both $\mathbf{R}_i$ and $\mathbf{R}_i^T$ are represented by 1×1 Conv. Normally, the initialization for $\mathbf{X}_i^0$ is acquired by upsampling the LrHSI. Here, we adopt the output of the previous diffusion time step as the initialized $\mathbf{X}_i^0$ for the input of P^{2}MSRN in the current step. This initialization strategy allows the network to benefit from the information already propagated through the diffusion process.

2) Invertible Denoising Network: Equation (8) can be equated with a denoiser based on Bayesian probability theory to counteract the noise in $\mathbf{X}_i^k$ and output the noise-free $\mathbf{G}_i^k$. While numerous deep networks (e.g., U-Net) have been introduced to achieve this goal, these models fail to maintain the detail and structure of information. The invertible denoising network (InDNet), characterized by a one-to-one mapping between input and output, represents a distinct paradigm within the field of neural networks. Its unique properties offer a range of advantages, including noise-free transfer and preservation of input details and structures. Besides, the same parameters are used for encoding and decoding, enabling the model light-weighted. Therefore, we choose InDNet as the denoiser, as shown in Fig. 3. For InDNet in each iteration, its internal primary workflow can be formalized as follows:

$$\psi^{\mathbf{H}*}[l+1, L] = \mathbf{H}[l+1, L] + \mathrm{F}_1(\mathbf{H}[1:l]) \tag{18}$$

$$\psi^{\mathbf{H}*}[1:l] = \mathbf{H}[1:l] \otimes \exp(\mathrm{F}_2(\psi^{\mathbf{H}*}[l+1:L])) + \mathrm{F}_3(\psi^{\mathbf{H}*}[l+1:L]) \tag{19}$$

$$\psi^{\mathbf{H}*}[1:L]\mathrm{cat} = \{\psi^{\mathbf{H}*}[1:l], \psi^{\mathbf{H}*}[l+1:L]\} \tag{20}$$

where $\psi^{\mathbf{H}*}[1:l]$ is the first to lth channels of the input, $\psi^{\mathbf{H}*}[l+1:L]$ denotes the input of the rest channels, $\otimes$ is the Hadamard product, and F_i $(i = 1, 2, 3)$ is bottleneck residual block.

Furthermore, InDNet takes on the mission of introducing the time step by encoding the time step using sine and cosine transformations and concatenating them in the last dimension. The concatenated representation passes through linear and activation layers to obtain the final time step representation, which will be attached with the input of InDNet as the embedding approach for the time step.

D. Cross-Scale Model-Driven Upsampling Network

After HSI is partly reconstructed on one specific consistent scale by P^{2}MSRN for certain time steps, the output is upsampled for the next level of restoration at a larger scale. However, mathematical interpolation can introduce nonuniform artifacts or distortions, which may adversely affect the overall image quality. Therefore, the CMUN is specifically designed based on the solution of (12) and (13) to increase the spatial resolution. Given the resemblance of the cross-scale upsampling model and scale-matched reconstruction model, the composition of CMUN and P^{2}MSRN is identical. Following the same reasoning, (12) can be unfolded into:

$$\begin{cases} \mathbf{Q}_{i_1}^k = \mathbf{X}_i^k \mathbf{S}_i - \mathbf{Z}_i \\ \mathbf{V}_{i_1}^k = \mathbf{Q}_{i_1}^k \mathbf{S}_i^T \end{cases} \tag{21}$$

$$\begin{cases} \mathbf{Q}_{i_2}^k = \mathbf{R}_i \mathbf{X}_i^k - \mathbf{Y}_i \\ \mathbf{V}_{i_2}^k = \mathbf{R}_i^T \mathbf{Q}_{i_2}^k \end{cases} \tag{22}$$

$$\mathbf{V}_i{}^k = \mathbf{X}_i{}^k - \mathbf{G}_i{}^k \tag{23}$$

$$\mathbf{X}_i{}^{k+1} = \mathbf{X}_i{}^k - \mu_2\left(\mathbf{V}_{i_1}^k + \mathbf{V}_{i_2}^k + 2\beta_2 \mathbf{V}_i{}^k\right). \tag{24}$$

Therefore, (21)–(24) correspond to the construction of TRNet, while (13) is applied in the design of InDNet.

In this case, $\mathbf{B}_i$ in TRNet should be changed to $\mathbf{S}_i$ accordingly so that the network can implement upsampling by passing $\mathbf{X}_\mathbf{i}^\mathbf{k}$ through CMUN branch. To the network structure, $\mathbf{S}_i$ is interpreted as a sequential consisting of 3×3 Conv, PReLU, and 2×2 Conv, and these two convolutional layers are for blurring and downsampling, respectively. And InDNet equated from (13) remains unchanged to function as the denoiser. The input to CMUN is made up of LrHSI $\mathbf{Z}$ and HrMSI $\mathbf{Y}$ with different resolutions, where the resolution of HrMSI is higher than that of HrMSI in the previous denoising stage, together with the result x_t of the last step. As a result, we can finally obtain a larger HSI for P^{2}MSRN to start super-resolving in the next phase.

Algorithm 1 Training

Input: HSI-SR network θ, scale factor f_t, noise schedule α, LrHSI $\mathbf{Z}$, HrMSI $\mathbf{Y}$, HrHSI $\mathbf{X}_0$

1: **repeat**
2: $\mathbf{X}_0 \sim q(\mathbf{X}_0)$
3: $t \sim \mathrm{Uniform}(1, \ldots, T)$
4: $\varepsilon \sim \mathcal{N}(0, I)$
5: $\mathbf{X}_t = \sqrt{\bar{\alpha}_t}(\mathbf{X}_0 \downarrow_{f_t}) + \sqrt{1-\bar{\alpha}_t}\varepsilon$
6: Optimize $||\mathbf{X}_0 - \mathbf{X}_\theta(\mathbf{Y}\uparrow_{1/f_t}, \mathbf{Z}\uparrow_{1/f_t}, t)||_1$
7: **until** converged

Algorithm 2 Sampling

Input: HSI-SR network θ, noise schedule α, scale factor f_t, LrHSI $\mathbf{Z}$, HrMSI $\mathbf{Y}$

1: $\mathbf{X}_T \sim N(0, \mathbf{I})$
2: **for** $t = T, \ldots, 1$ **do**
3: $(\mu, \theta) \sim \mathcal{N}(0, I)$ if $t > 1$, else $(\mu, \theta) = 0$
4: Sample the denoising output by Eq.(2)
5: **end for**
6: **return** $\mathbf{X}_0$

E. Training Objective

According to DDRF [37], predicting the original data directly is more efficient than noise, due to the small-scale datasets for the HSI-SR task. Therefore, we choose to predict x_0 as well. The general flow of training is outlined

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in Algorithm 1. Besides, Algorithm 2 shows the specific sampling schedule of the proposed ISPDiff. In the experiment, we choose L_1 as the training loss function, which is regarded as the preferred option for HSI-SR owing to its capability to preserve image edges and texture structures. The loss function can be expressed as follows:

$$\mathcal{L} = \mathbb{E}[||\mathbf{X}_0 - \mathbf{X}_\theta(\mathbf{Y}\uparrow_{1/f_t}, \mathbf{Z}\uparrow_{1/f_t}, t)||_1] \tag{25}$$

where $\mathbf{X}_0$ is the ground truth, $\mathbf{X}_\theta$ is the HrHSI predicted by the network conditioned on HrMSI $\mathbf{Y}$, and LrHSI $\mathbf{Z}$ with scale factor f_t, and time step t.

IV. Experiments

In this section, we validate the performance of the proposed ISPDiff through comparisons with various competing algorithms on three simulated datasets and one real dataset. Both quantitative and qualitative results of all methods are presented in this section. Then, we conduct several ablation experiments to assess the effectiveness of different modules and strategies within ISPDiff.

A. Datasets and Evaluation Metrics

We test our method on three public datasets, CAVE [38], Pavia Center [39], and Harvard [40] datasets.

1) CAVE Dataset: The CAVE dataset was obtained by using a generalized assorted pixel camera, containing 32 high-quality indoor HSIs. The size of each HSI is 512 × 512 with 31 bands which range from 400 to 700 nm. In the experiments, we randomly choose 22 HSIs as training samples while the rest is regarded as testing samples. Besides, Wald's protocol [41] is applied to HSIs generate HrMSI and LrHSI. To obtain more training samples, HrMSIs are cropped into patched with the size of 128 × 128 × 3, while the size of LrHSI patches is 32 × 32 × 31.

2) Pavia Center Dataset: The Pavia Center dataset, acquired by the reflective optics system imaging spectrometer (ROSIS) sensor, contains 115 bands ranging from 430 to 860 nm. This dataset is represented as an image with dimensions of 1096 × 1096 pixels, and it exhibits a geometric resolution of 1.3 m. We excluded 13 bands with noise, resulting in a dataset for experimentation consisting of 102 bands. Furthermore, we eliminated a region of the image exhibiting poor quality, ultimately utilizing the processed image with dimensions 960 × 640 × 102 as a reference. This reference image is partitioned into 12 nonoverlapping patches sized 160 × 320 × 102, with nine patches designated for training and three for testing. The training dataset is further subdivided into 21 overlapping smaller patches, each measuring 160 × 160 × 102. Similarly, LrHS images with dimensions 40 × 40 × 102 pixels and PAN images sized 160 × 160 are generated following Wald's protocol.

3) Harvard Dataset: The Harvard dataset provides a total of 50 images under daylight illumination, both indoors and outdoors, using a commercial HS camera with a spatial resolution of 1040 × 1392 and a series of 31 bands ranging from 420 to 720 nm. Following Wald's protocol, HrMSIs in the Harvard dataset are cropped into patches with the size of 320 × 320 × 3 in the training set, and the size of LrHSIs is 80 × 80 × 31. In addition, in the testing set, HrMSIs are used with the original size and LrHSIs are 260 × 348 × 31. For numerical stability, each patch in three datasets is normalized by dividing the maximum value to make the pixels range from 0 to 1.

4) Real Dataset: To assess the practical effectiveness of the proposed method, we collect a dataset in the Chang'an District of Xi'an, Shaanxi, China. This real dataset comprises an HSI with the size of 960 × 960 × 137, collected by the Headwall VNIR/SWIR Full-band Hyperspectrometer, and an MSI measuring 3840 × 3840 × 3, obtained from the FLIR Duo Pro R sensor. As ground truth is unavailable for real-world data, we utilized Wald's protocol to construct the training set for supervised training. The obtained HSI serves as the reference to generate lower spatial resolution HSIs sized at 240 × 240 × 137 and lower spectral resolution MSIs at 960 × 960 × 3. Training samples are created with two lower-resolution images mentioned above through overlapping cropping, resulting in 69 pairs of HSI and MSI each sized at 40 × 40 × 137 and 160 × 160 × 3, respectively. While the network is adequately trained, real-world LrHSI and HrMSI are utilized for testing. The test set comprises 69 real image pairs created in the same way, sized at 160 × 160 × 137 and 640 × 640 × 3.

5) Evaluation Metrics: Various established image quality assessment metrics are employed to comprehensively evaluate the performance. These metrics encompass a peak signal-to-noise ratio (PSNR), which measures the fidelity of the reconstructed image; correlation coefficients (CCs), indicating the linear relationship between the reference and reconstructed images; spectral angle mapper (SAM), quantifying spectral similarity; root mean square error (RMSE), providing an overall measure of the reconstruction error; and erreur relative globale adimensionnelle de synthèse (ERGAS), a metric of overall fusion quality that measures mean shifting and dynamic range change. In the absence of ground truth for the real dataset, we employ the quality w/no reference QNR = $(1 - D_\lambda)^\alpha (1 - D_S)^\beta$ [42] for evaluation, where D_λ and D_S assesses the spectral and spatial distortions, respectively. QNR provides a comprehensive index of both aspects, while QNR goes higher, D_λ and D_S remain lower, the quality of the fused results gets better. The inclusion of this diverse set of metrics ensures a thorough efficacy evaluation of the competing methods and our proposed method across various dimensions of image quality.

B. Comparison With SOTA Methods

To evaluate the effectiveness of our proposed method, a series of comprehensive experiments are conducted between our method and state-of-the-art HSI-SR methods. To be specific, five DL based algorithms are chosen for comparison, namely, UAL [43], MoG-DCN [44], LAGConv [45], SFPN [15], PSRT [46], and DDPM-Fus [47]. In addition, we benchmark our method against several classic methods as well, including, GSA [48], CNMF [21], and HySure [49].

C. Implementation Details

The proposed ISPDiff is implemented in the PyTorch framework with one NVIDIA GeForce GTX 3090. In the

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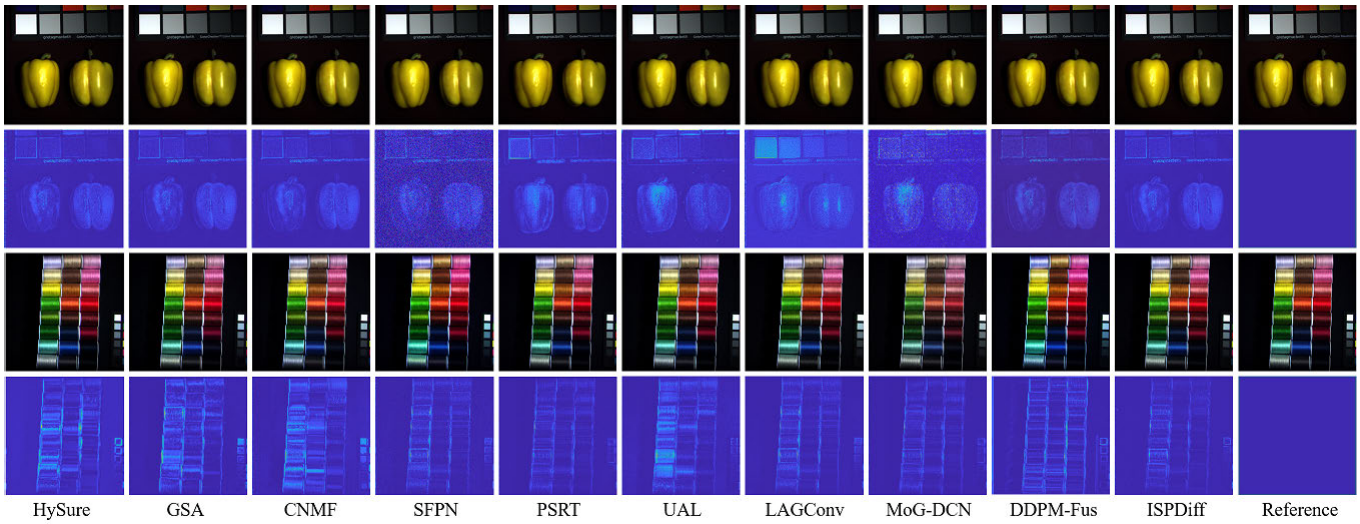


Fig. 4. Visual quality comparison of all comparison methods on the CAVE dataset, and the error maps highlight the differences between the SR results and the ground truth.

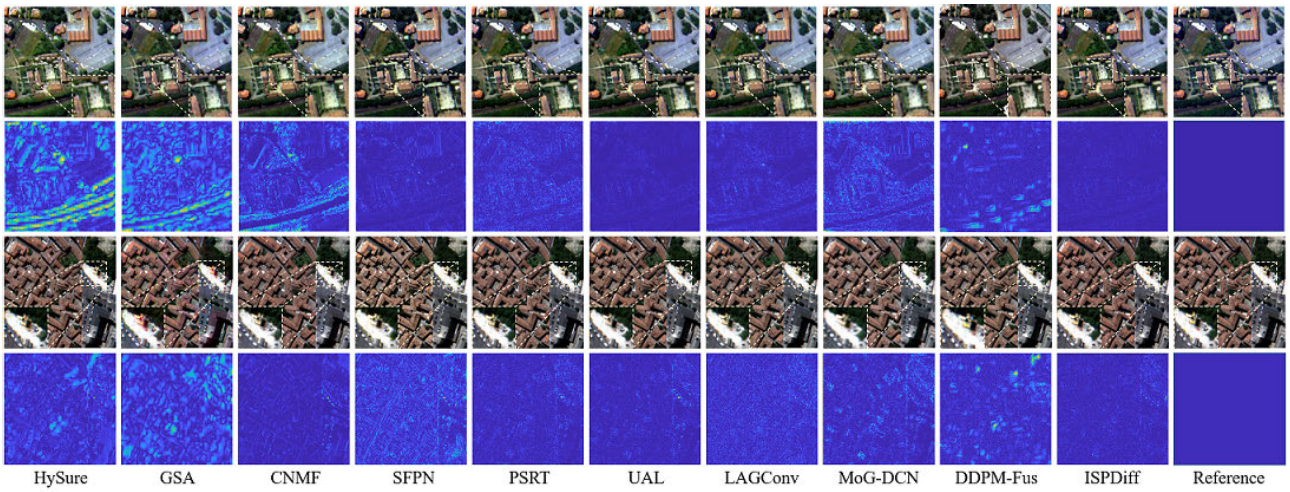


Fig. 5. Visual quality comparison of all comparison methods on the Pavia Center dataset, and the error maps highlight the differences between the SR results and the ground truth.

training phase, we set the batchsize as 4 and the model is optimized by Adam optimizer over 1500 epochs with a learning rate of 10^{-4}. Indeed, the network can be applied to 2^i times HSI-SR, and then the backward process consists of $i+1$ scales. To be specific, we set the downsampling factor f_t to $[2^0, \ldots, 2^0, 2^{-1}, \ldots, 2^{-1}, \ldots, 2^{-i}, \ldots, 2^{-i}]$. For the diffusion process, we choose a cosine schedule for α_t, and set $s = 8e^{-3}$, total time steps $T = 1500$.

D. Results and Analysis

1) Qualitative Comparison: We select the 10th, 20th, and 29th bands to visualize the results of reconstruction obtained by all the competing methods tested on the yellow peppers and thread spools images from the CAVE dataset in Fig. 4. To enhance clarity, we incorporate error maps at the 27th band, highlighting differences between each result and the reference, given the challenge of human visual discrimination. While the classical methods show great potential in reconstructing the yellow peppers image with fewer artifacts, a notable deviation is observed in the edges and lines of the spread spools image, as evident in the error map. Contrastingly, both SFPN and PSRT yield results with varying levels of noise, signaling a constrained improvement in noise resistance. The results of DDPM-Fus have a relatively obvious spectral deviation. Intriguingly, our proposed ISPDiff method outshines the comparative techniques, showcasing remarkable performance with minimal spatial and spectral distortions. This is particularly evident when compared with the results from UAL, LAGConv, and MoG-DCN, where notable deviations are observed, reinforcing the superior capabilities of ISPDiff in achieving high-fidelity spectral and spatial reconstructions. The robust generalization ability of our model stems from its inherent explicability, allowing for a clear understanding of the internal implementation mechanism. This transparency in the model's structure and operations empowers us to discern and comprehend its decision-making processes, ultimately contributing to its superior performance across diverse scenarios.

The visual results of the competing algorithms on two patches cropped form the Pavia Center dataset are illustrated in Fig. 5. We choose the 5th, 35th, and 55th bands and sole 70th bands to visualize the pseudo-color images and error

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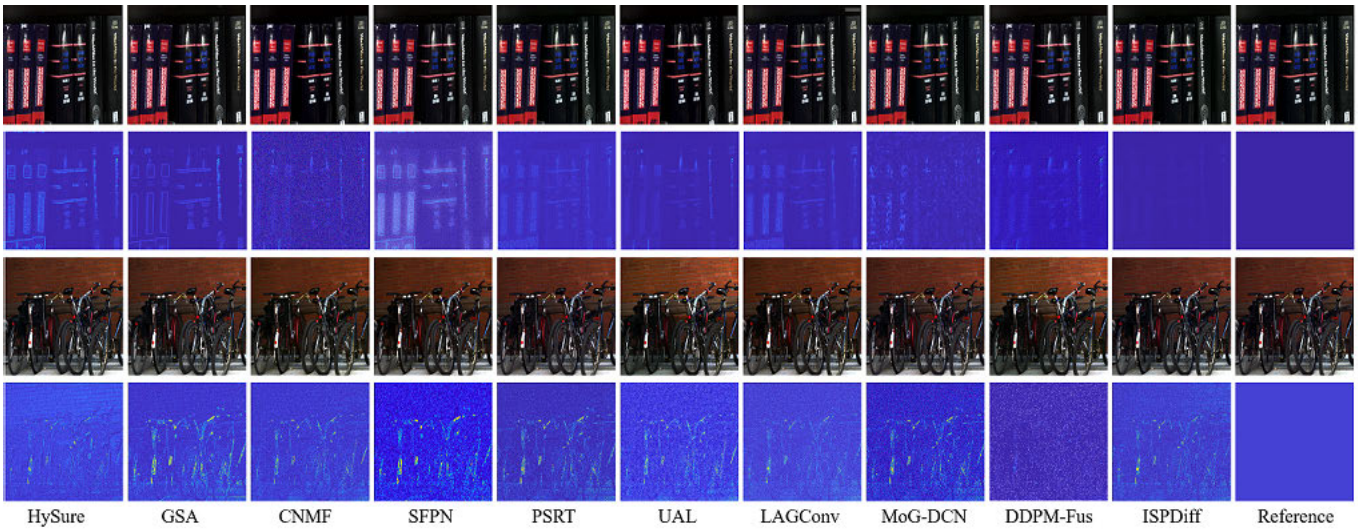


Fig. 6. Visual quality comparison for results of all competing methods on the Harvard dataset, and the error maps highlight the differences between the SR results and the ground truth.

maps, respectively. The classical algorithms exhibit subpar performance in reconstructing fine details as shown in pseudo-color images, as evidenced by a significant deviation between the SR results and the ground truth indicated by the error maps. In contrast, DL-based methods showcase commendable results. However, upon inspection through a close-up view, it becomes apparent that SFPN, PSRT, UAL, LAGConv, and DDPM-Fus exhibit noticeable spectral distortion despite their impressive reconstruction. MoG-DCN, also an interpretable approach, demonstrates shortcomings in local detail recovery, manifesting a blurring effect. Notably, the proposed ISPDiff emerges as the frontrunner in achieving the closest reconstruction fidelity to the reference, which underscores the effectiveness of our approach in preserving intricate details and maintaining a high level of visual consistency in comparison to both classical and DL-based methods.

The visualized results at the combination of the 10th, 20th, and 29th bands of all the comparison methods on the Harvard dataset are shown in Fig. 6. Besides, we take the 27th band to plot the error map. Similar to the outcomes from the aforementioned datasets, traditional methods still exhibit unsatisfactory reconstruction. Notably, these methods manifest significant residual, particularly evident in the edge regions of the image. Results obtained by SFPN display a considerable color disparity compared to the ground truth, accompanied by pronounced artifacts. Simultaneously, the outcomes obtained by PSRT, UAL, LAGConv, MoG-DCN, and DDPM-Fus reveal levels of spatial and spectral distortions introducing the overall quality of the reconstruction, which contributes to the suboptimal performance of reconstruction. In contrast, the proposed ISPDiff distinguishes itself by presenting fewer artifacts when compared to its counterparts. This highlights the efficacy of ISPDiff in preserving image quality, demonstrating a notable reduction in reconstruction errors. The balance achieved by ISPDiff between fidelity and clarity underscores its potential as an advanced and reliable solution in the realm of image reconstruction.

TABLE I

QUANTITATIVE PERFORMANCE OF DIFFERENT METHODS ON THE CAVE DATASET. THE BEST AND THE SECOND BEST VALUES ARE HIGHLIGHTED IN **BOLD** AND UNDERLINED, RESPECTIVELY

Methods	PSNR↑	CC↑	SAM↓	RMSE↓	ERGAS↓
HySure	37.2266	0.9940	5.5511	0.0140	3.5572
GSA	39.3864	0.9963	4.9586	0.0111	2.7519
CNMF	38.2289	0.9941	4.4831	0.0127	3.2748
UAL	42.1837	0.9983	3.2864	0.0083	1.8982
LAGConv	45.3103	0.9989	3.1348	0.0057	1.3991
MoG-DCN	44.3221	0.9989	2.6550	0.0065	1.4273
SFPN	44.6061	0.9979	2.7518	0.0064	1.4348
PSRT	43.1880	0.9985	2.8130	0.0075	1.6794
DDPM-Fus	42.7024	0.9896	3.2552	0.0089	1.8068
Proposed	**46.2880**	**0.9990**	**2.6469**	**0.0038**	**1.3981**

We take two patches from the real dataset to visualize the 22nd, 34th, and 46th bands of the fused HSIs reconstructed from all methods in Fig. 7. While these three traditional methods exhibit good spatial fidelity, they also manifest some degree of spectral distortion akin to SFPN and PSRT. The results of UAL and DDPM-Fus retain some noise, while artifacts are noticeable in LAGConv and MoG-DCN. The proposed method demonstrates favorable reconstruction in both spatial and spectral aspects.

2) Quantitative Comparison: To validate the superior performance of ISPDiff, Tables I–III and Fig. 8 provide a comprehensive overview of quantitative results among the nine discussed methods and our proposed method across four datasets at an upsampling ratio of 4. It should be noted that the performance metrics are averaged across all testing samples, comprising ten samples for CAVE and Harvard, 63 samples

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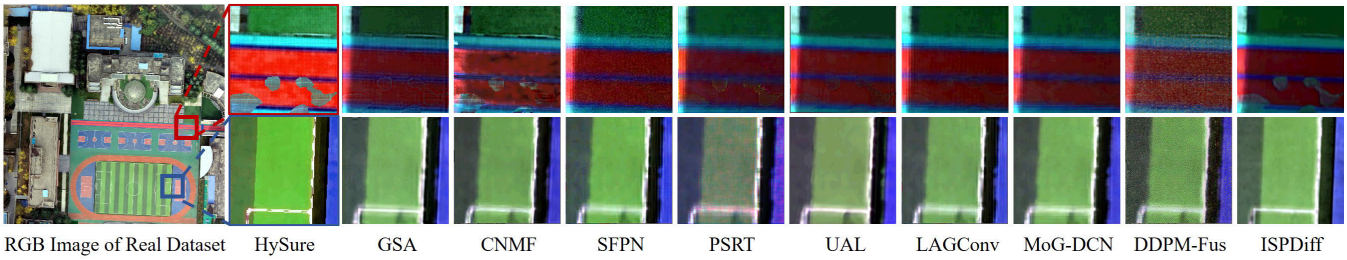


Fig. 7. Visual quality comparison for results of all competing methods on a real dataset.

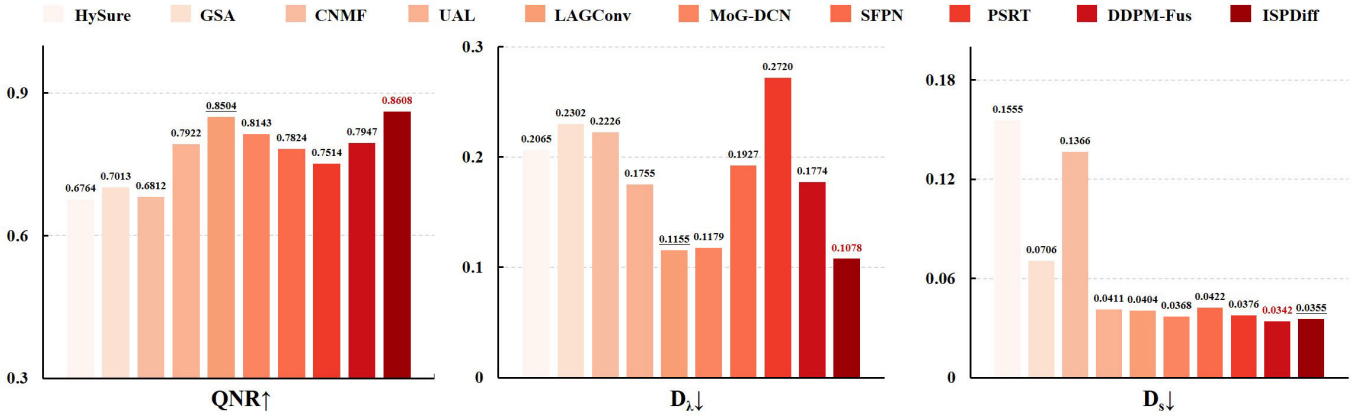


Fig. 8. Quantitative performance of different methods on the real dataset, where the best and the second best are highlighted in red and underlined, respectively.

TABLE II

QUANTITATIVE PERFORMANCE OF DIFFERENT METHODS ON THE PAVIA CENTER DATASET. THE BEST AND THE SECOND BEST VALUES ARE HIGHLIGHTED IN **BOLD** AND UNDERLINED, RESPECTIVELY

Methods	PSNR↑	CC↑	SAM↓	RMSE↓	ERGAS↓
HySure	33.6376	0.9658	7.1436	0.0211	3.6249
GSA	33.1199	0.9672	5.9771	0.0222	3.3465
CNMF	31.8147	0.9535	4.9681	0.0262	4.1838
UAL	40.9412	0.9868	3.3791	0.0091	2.0273
LAGConv	41.4828	0.9884	3.2784	0.0085	1.8738
MoG-DCN	42.9636	0.9907	2.7681	0.0072	1.5435
SFPN	41.9923	0.9901	2.9615	0.0080	1.7726
PSRT	40.1077	0.9827	3.0693	0.0113	2.2747
DDPM-Fus	42.4565	0.9903	3.0030	0.0076	1.6605
Proposed	**44.4740**	**0.9923**	**2.5920**	**0.0060**	**1.4234**

TABLE III

QUANTITATIVE PERFORMANCE OF DIFFERENT METHODS ON THE HARVARD DATASET. THE BEST AND THE SECOND BEST VALUES ARE HIGHLIGHTED IN **BOLD** AND UNDERLINED, RESPECTIVELY

Methods	PSNR↑	CC↑	SAM↓	RMSE↓	ERGAS↓
HySure	44.0898	0.9860	4.0137	0.0069	3.2131
GSA	38.7696	0.9769	3.4038	0.0122	4.1272
CNMF	45.5694	0.9908	3.2340	0.0059	2.3554
UAL	46.4193	0.9847	3.2414	0.0050	2.3661
LAGConv	48.1992	0.9918	3.6260	0.0041	1.9903
MoG-DCN	47.0182	0.9881	3.3882	0.0047	2.3567
SFPN	46.9662	0.9902	3.4624	0.0049	2.2106
PSRT	46.8713	0.9867	3.3597	0.0048	2.4541
DDPM-Fus	45.0946	0.9872	3.4336	0.0090	2.0001
Proposed	**49.6497**	**0.9945**	**2.8437**	**0.0034**	**1.7583**

for Pavia Center, and 69 samples for the real dataset. It can be inferred from the results that ISPDiff outperforms all competing methods by a significant margin across all evaluation metrics, particularly in PSNR and RMSE, confirming its excellence in HSI-SR. As for the real dataset, Fig. 8 illustrates that ISPDiff reaches a pleasant tradeoff in the reconstruction of spatial and spectral domains, inferred from the comparison of QNR. This remarkable performance of ISPDiff is attributed to the interpretable multiscale reconstruction, allowing it to refine spatial details while preserving spectral information. Additionally, our proposed method achieves a more favorable balance between performance and model size, as illustrated in Fig. 9. The depicted comparison demonstrates that these competitors exhibit a slight inferiority to our method in terms of Params and FLOPs.

E. Ablation Study

1) Effect of Scale-Propelled Diffusion: The experiment in this section explores the effectiveness of scale-propelled dif-

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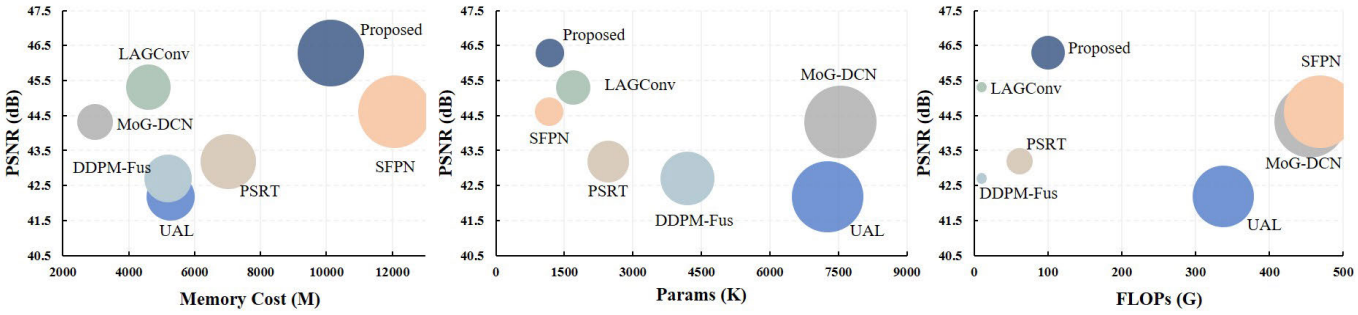


Fig. 9. PSNR performance of DL-based methods versus memory cost, number of parameters, and number of FLOPs.

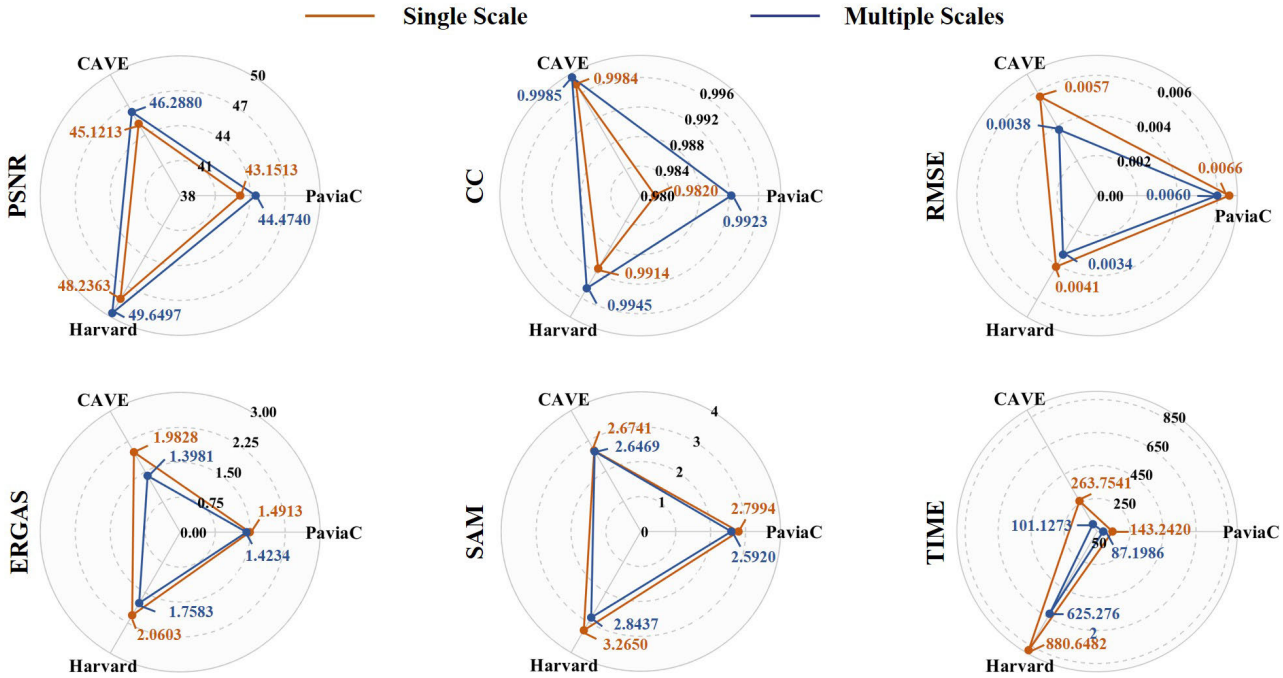


Fig. 10. Ablation experiments on the effect of the proposed method with single scale and multiple scales on the CAVE, Pavia Center, and Harvard datasets.

fusion. The promising results substantiate the pivotal role of the multi-scale strategy in ISPDiff, affirming its significance in achieving superior results. Through the verification on three distinct datasets, it becomes evident that the single-scale approach for HSI-SR falls short in comparison to the multi-scale effect, as shown in Fig. 10. Furthermore, the multi-scale ISPDiff exhibits a prominent shorter average inference time, effectively addressing the problem of the time-consuming inference in DDPM. Thus, the incorporation of the multi-scale strategy not only enhances the performance but also mitigates the computational burden. The effectiveness of this scheme stems from its iterative refinement steps, which enable the capture of richer information on the data distributions of HrHSIs from several scales.

2) Effect of P^{2}MSRN and CMUN in ISPDiff: To comprehensively evaluate the effectiveness of the network architecture proposed in this article, we conduct an extensive ablation experiment. In this section, we design a variant of the proposed model(Variant-w/U-Net), in which P^{2}MSRN and CMUN in ISPDiff are replaced with U-Net. By analyzing objective indicators across three diverse datasets, we observe a significant superiority of ISPDiff over U-Net, as shown in Table IV. This superiority can be attributed to the theoretical derivation and integration of data characteristics within ISPDiff, thereby providing it with a deeper level of physical significance compared to the inherent "black-box" nature of U-Net.

3) Effect of Number of Stages in P^{2}MSRN: To thoroughly evaluate the impact of the numbers of stages in P^{2}MSRN, we conduct a performance comparison by training and testing four variants of the model with varying numbers of stages (specifically, 2, 3, 4, and 5 stages). The results, as shown in Table V, clearly demonstrate how the SR performance acts as the number of stages increases from 2 to 5. It is worth noting that further increasing the number of stages tends to yield diminishing quality when reaching a certain threshold. One potential explanation is that, in this experiment, the proposed model achieved convergence after three iterations. Therefore, for the sake of better performance, we choose the number of stages to be 3.

4) Effect of InDNet: The denoiser plays a crucial role in establishing a more robust network and attaining superior reconstruction performance. To illustrate the efficacy of InD-

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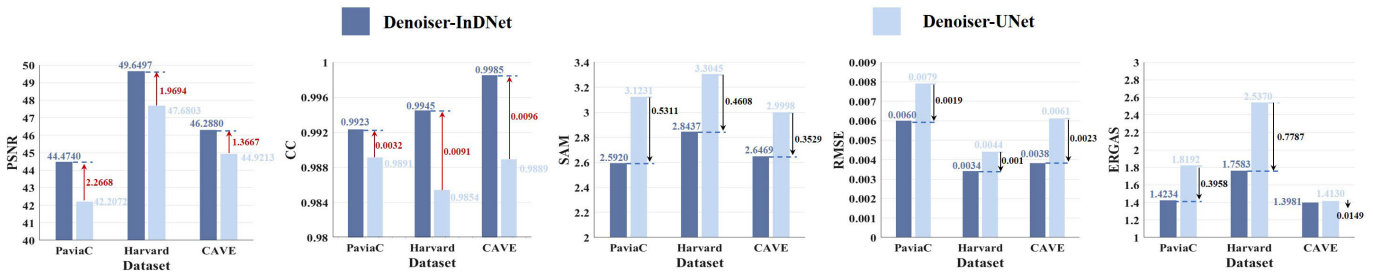


Fig. 11. Ablation experiments on the effects of different denoisers in P^{2}MSRN and CMUN on the Pavia Center, CAVE and Harvard datasets.

TABLE IV

ABLATION EXPERIMENTS ON THE EFFECT OF P^{2}MSRN AND CMUN IN ISPDIFF ON THREE DATASETS

Datasets	Model	PSNR↑	CC↑	SAM↓	RMSE↓	ERGAS↓
Pavia	ISPDiff	**44.4740**	**0.9923**	**2.5920**	**0.0060**	**1.4234**
	Variant-w/U-Net	36.4072	0.9860	5.2852	0.0156	2.8642
CAVE	ISPDiff	**46.2880**	**0.9985**	**2.6469**	**0.0038**	**1.3981**
	Variant-w/U-Net	33.0010	0.9872	3.9773	0.0237	5.2661
Harvard	ISPDiff	**49.6497**	**0.9945**	**2.8437**	**0.0034**	**1.7583**
	Variant-w/U-Net	38.1761	0.9809	3.2171	0.0051	3.0438

TABLE V

ABLATION EXPERIMENTS ON THE EFFECT OF THE NUMBER OF STAGES IN P^{2}MSRN ON THE PAVIA CENTER AND CAVE DATASETS

Datasets	Stages	PSNR↑	CC↑	SAM↓	RMSE↓	ERGAS↓
PaviaC	2	43.3212	0.9914	2.7700	0.0062	1.4422
	3	**44.4740**	**0.9923**	**2.5920**	**0.0060**	**1.4234**
	4	43.7122	0.9916	2.6654	0.0066	1.5275
	5	43.3933	0.9913	2.7413	0.0069	1.5773
CAVE	2	45.2681	0.9981	2.7491	0.0058	1.4597
	3	**46.2880**	**0.9985**	**2.6469**	**0.0038**	**1.3981**
	4	45.3331	0.9984	2.6952	0.0057	1.4228
	5	45.2998	0.9984	2.7017	0.0057	1.4408

Net as the denoiser, we conduct a comparative experiment between our proposed model and its variant, where we replace InDNet with the widely-used U-Net while keeping other training settings fixed. It can be seen from Fig. 11 that our proposed model with InDNet denoiser outperforms that with U-Net denoiser in terms of five commonly used objective indexes, indicating that InDNet is more capable of retaining detail and structure information.

5) Effect of Different Diffusion Frameworks: To explore the performance of our proposed method under different diffusion frameworks, we embed the proposed approach in the DDIM [50] and I-DDPM [51] frameworks and make evaluations on the Pavia Center dataset. The results in Table VI reveal that although DDIM and I-DDPM yield satisfactory outcomes at 100 sampling steps, they fall short of the performance achieved at 1500 steps. Moreover, there exists a discernible

TABLE VI

EXPERIMENTS ON THE EFFECT OF DIFFERENT DIFFUSION FRAMEWORKS ON THE PAVIA CENTER DATASET

Framework	Steps	PSNR↑	CC↑	SAM↓	RMSE↓	ERGAS↓
DDIM	100	41.8680	0.9890	2.6469	0.0065	1.4939
	1500	43.1530	0.9919	2.6213	0.0062	1.4611
I-DDPM	100	42.3415	0.9900	2.7908	0.0069	1.5614
	1500	44.2240	0.9914	2.7567	0.0066	1.5265
DDPM	1500	**44.4740**	**0.9923**	**2.5920**	**0.0060**	**1.4234**

disparity between the optimal results of both methods and those of DDPM. This discrepancy can be attributed to the primary focus of I-DDPM and DDIM on expediting the sampling process and improving generation efficiency. However, these improvements may not be optimally suited for addressing the intricacies and detailed requirements inherent in HSIs. Given that HSI data entail significantly higher complexity compared to conventional RGB images due to their multitude of spectral channels, I-DDPM and DDIM might necessitate more nuanced adjustments and specially tailored architectures to effectively handle such high-dimensional data.

V. CONCLUSION

In this article, we present a novel ISPDiff model called ISPDiff, especially for HSI-SR. Specifically, ISPDiff follows a pyramid-like process. While destroying and downsampling HrHSI with Gaussian noise in the scale-diminished diffusion process, we design a transparent unified scale-flexible model in the scale-propelled denoising process based on the physical principles of HSI-SR. Then we solve this model with iterative optimization algorithms which are one-to-one unfolded into two distinct networks: P^{2}MSRN for scale-matched restoration and CMUN for cross-scale upsampling. The corrupted LrHSI is then progressively restored by P^{2}MSRN for some steps with HrMSI as a condition and upsampled to a finer level by CMUN, and so forth. This ISPDiff scheme significantly enhances the explainability and effectively reduces the inference time. Systematic experimental investigations have been conducted on three public datasets to validate that the proposed method outperforms other state-of-the-art methods both quantitatively and qualitatively. In our future work, we intent to extend the proposed method to incorporate various data sources, including PAN, SAR, and LiDAR, to improve the

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adaptability and effectiveness of ISPDiff across diverse AI domains such as environmental monitoring and medical image processing. This comprehensive approach aims to make significant contributions to the advancement of AI technology in multisource data fusion.

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